Strategy-Relative Market Regimes as Filtration Compressions

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Abstract

Classical notions of market regimes are defined at the level of the underlying price process. We propose a different formalisation: regimes are defined relative to a given trading strategy through the conditional law of future profit and loss (PnL). We model regimes as adapted binary processes that compress the filtration and separate the distribution of future PnL while remaining temporally persistent. We formulate this as a stochastic control problem with switching costs, prove existence and uniqueness of optimal regime processes under mild conditions, and interpret regimes as minimal predictive state variables for strategy performance. Connections to information-theoretic formulations are discussed.

1 Introduction

Market regimes are typically defined as latent or observable states governing the dynamics of an underlying price process. In such approaches, the regime is a property of the market itself, independent of any trading strategy.

However, in practice, the relevant object is not the behaviour of the price process but the behaviour of *strategy PnL*. The same market dynamics may be favourable for one strategy and unfavourable for another. This suggests that regimes should be defined relative to a strategy rather than intrinsically at the level of the market.

We propose a formalisation in which a regime is an adapted binary process that summarises available information and separates the conditional distribution of future PnL. The regime is thus a *compression of the filtration* that preserves predictive information relevant for trading performance.

2 Set-up

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0}^T, \mathbb{P})$ be a filtered probability space.

Let Z_t be an $\mathcal{F}_t$-adapted process taking values in a Polish space E. This represents the observable state (e.g. prices and derived features).

Let X_{t+1} denote the one-step PnL increment of a fixed strategy:

$$X_{t+1} := \Pi_{t+1} - \Pi_t,$$

with a regular conditional law given Z_t.

Definition 1 (Regime process)**.** *A regime process is a binary adapted process*

$$s_t \in \{0, 1\}, \qquad s_t \in \sigma(Z_0, \ldots, Z_t, s_{t-1}).$$

The interpretation is that $s_t = 1$ corresponds to a favourable regime (strategy active), and $s_t = 0$ to an unfavourable regime.

3 Objective

Let $u : \mathbb{R} \to \mathbb{R}$ be a bounded measurable utility function, and let $\lambda \geq 0$ be a switching cost.

We define the objective functional

$$J(s) = \mathbb{E}\left[\sum_{t=0}^{T-1} \left(u(s_t X_{t+1}) - \lambda \mathbf{1}_{\{s_t \neq s_{t-1}\}}\right) + h(Z_T, s_{T-1})\right],$$

where h is a bounded terminal reward.

Definition 2 (Optimal regime process)**.** *A regime process s^* is optimal if it maximises $J(s)$ over all admissible regime processes.*

4 Dynamic Programming Formulation

Define the augmented state

$$Y_t := (Z_t, s_{t-1}) \in E \times \{0, 1\}.$$

Define

$$m_t(z, a) := \mathbb{E}[u(aX_{t+1}) \mid Z_t = z].$$

The reward function is

$$r_t(z, a, a^-) = m_t(z, a) - \lambda \mathbf{1}_{\{a \neq a^-\}}.$$

5 Existence

Theorem 1 (Existence and Markov structure)**.** *Assume:*

- *Z_t is Markov with transition kernel $P_t(\cdot \mid z)$,*
- *$m_t(z, a)$ is bounded and Borel measurable,*
- *h is bounded and Borel measurable.*

Then there exists an optimal regime process s^*. Moreover, it admits a Markov feedback representation

$$s_t^* = \pi_t^*(Z_t, s_{t-1}^*)$$

for some measurable selector π_t^*.

Proof. Define the Bellman recursion:

$$V_T(z, a^-) = h(z, a^-),$$

$$V_t(z, a^-) = \max_{a \in \{0,1\}} \left\{ r_t(z, a, a^-) + \int_E V_{t+1}(z', a)\, P_t(dz' \mid z) \right\}.$$

Since the action space is finite and all functions are measurable and bounded, the maximum exists pointwise and measurable selectors exist. Standard backward induction yields optimality. □

6 Uniqueness

Define the action-value difference

$$\Delta_t(z, a^-) := Q_t(z, a^-, 1) - Q_t(z, a^-, 0).$$

Theorem 2 (Uniqueness under strict separation)**.** *If for each t,*

$$\mathbb{P}\big(\Delta_t(Z_t, s^*_{t-1}) = 0\big) = 0,$$

then the optimal regime process is unique up to indistinguishability.

Proof. At each time t, optimality requires choosing an action maximising Q_t. Under the no-tie condition, this maximiser is unique almost surely. Backward induction implies uniqueness of the entire process. □

7 Characterisation

The optimal regime satisfies

$$s^*_t = \mathbf{1}_{\{\Delta_t(Z_t, s^*_{t-1})>0\}}.$$

Thus the regime is a threshold rule based on a dynamic advantage function incorporating:

- immediate expected PnL,
- switching cost,
- continuation value.

8 Interpretation as Filtration Compression

The regime process can be written as

$$s_t = f_t(Z_t, s_{t-1}),$$

i.e. a binary compression of the filtration.

The optimal regime balances:

- predictive power for future PnL,
- temporal persistence,
- simplicity of representation.

Remark 1. *Under this interpretation, regimes are not intrinsic properties of the market, but minimal predictive state variables for a given strategy.*

9 Information-Theoretic Perspective

An alternative formulation considers

$$\max_s \quad I(s_t; X_{t+1}) - \lambda \mathbb{E}[\mathbf{1}_{\{s_t \neq s_{t-1}\}}],$$

or more generally

$$\max_f \quad I(f(Z_t); X_{t+1}) - \beta I(f(Z_t); Z_t) - \lambda \mathbb{E}[\mathbf{1}_{\{f(Z_t) \neq f(Z_{t-1})\}}],$$

which corresponds to a temporally regularised information bottleneck.

The control formulation may be viewed as a utility-weighted version of this objective.

10 Conclusion

We have proposed a definition of market regimes as adapted binary processes that compress the filtration while preserving predictive information about future strategy PnL. The resulting framework yields existence and uniqueness results, admits a dynamic programming characterisation, and provides a natural bridge between stochastic control and information-theoretic approaches to regime detection.